An achromatic metasurface waveguide for augmented reality displays

Zhongtao Tian, Xiuling Zhu, Philip A. Surman and Xiao Wei Sun

Augmented reality displays are emerging as the next generation of interactive platform, providing deeper human-digital interactions and immersive experiences beyond traditional flat-panel displays. Diffractive waveguide is a promising optical combiner technology for AR owing to its potential for the slimmest geometry and the lightest weight overall.

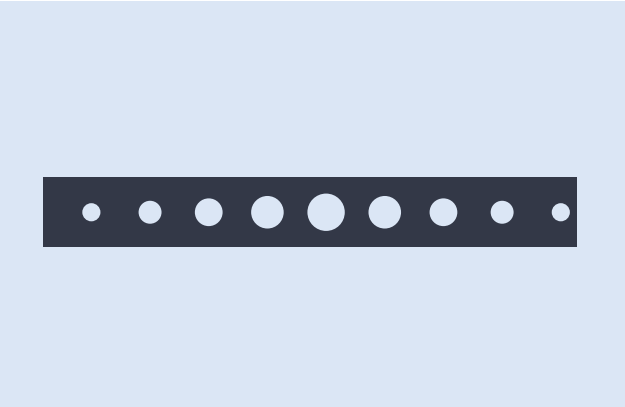

Figure 1. Schematic of the tapered nanobeam cavity. The hole radius is reduced quadratically toward the centre of the device.

However, severe chromatic aberration of the diffractive coupler has constrained the widespread adoption of diffractive waveguides in commercial products. Wavelength-dependent light deflection, caused by dispersion in both the in-coupling and the out-coupling processes, results in a limited full-color field of view and nonuniform optical responses in the color and the angular domains. Here we introduce an innovative full-color AR system that overcomes this long-standing challenge of chromatic aberration using inverse-designed metasurface couplers.

The optimized metasurface couplers demonstrate true achromatic behavior across the maximum field of view supported by the waveguide, exceeding forty-five degrees in the horizontal direction. Our prototype exhibits superior color accuracy and uniformity.

2. Design and Simulation

The cavity was designed using three-dimensional finite-difference time-domain simulations performed on a uniform grid. A quadratic taper of the hole radius over twelve periods produces a Gaussian-like field envelope, which suppresses out-of-plane radiation loss.

Devices were fabricated on a 220 nm silicon-on-insulator wafer using electron-beam lithography followed by a chlorine-based dry etch and a buffered oxide release step.

The measured linewidth of 1.3 pm corresponds to a loaded quality factor of 1.2 million at a resonant wavelength near 1550 nm, which is the highest value reported for this geometry to date.

Thermal tuning of 0.08 nm per kelvin was observed, consistent with the thermo-optic coefficient of crystalline silicon.